Project Management Handbook

Capulus Technologies - Project Management Handbook
Version 3.0 - Effective January 1, 2024

1.0 Introduction
This handbook defines project management standards for all Capulus projects, with special emphasis on AI■driven solutions for smart policing, smart cities, and vision AI.

2.0 Project Methodology
2.1 Methodology Selection
Project Type | Recommended Methodology

| ---------- | ---------- |
Software Development (AI) | Agile/Scrum

Infrastructure (Smart City) | Waterfall with gates

Marketing Campaign | Hybrid

Research (Vision AI) | Kanban

Enterprise■wide | SAFe

2.2 Agile Framework
Sprint Structure:
• Duration: 2 weeks (fixed).
• Ceremonies:
• Sprint Planning: 2 hours (Day 1)
• Daily Stand■up: 15 minutes (9:30 AM)
• Sprint Review: 1 hour (Last day)
• Retrospective: 1 hour (Last day)
• Backlog Refinement: 1 hour/week

Roles:
• Product Owner: Defines requirements, prioritizes backlog.
• Scrum Master: Facilitates process, removes blockers.
• Development Team: 5-7 members, cross■functional.
• Stakeholders: Provide feedback at reviews.

3.0 Project Documentation
3.1 Required Documents by Phase
Initiation:
• Project Charter (mandatory)
• Business Case
• Stakeholder Register

Planning:
• Project Plan (mandatory)
• Risk Register (mandatory)
• Communication Plan
• Budget Approval

Execution:
• Status Reports (weekly)
• Issue Log
• Change Requests
• Meeting Minutes

Closure:
• Lessons Learned (mandatory)

• Final Report
• Handover Documents

4.0 Project Metrics and KPIs
4.1 Mandatory Metrics
• Schedule Variance (SV)
• Cost Variance (CV)
• ROI/TCO Analysis
• Resource Utilization
• Quality Metrics (defect rate)

4.2 Agile Metrics
• Velocity
• Sprint Burndown
• Release Burndown
• Cumulative Flow
• Cycle Time

4.3 Success Criteria
Criterion | Target

| ---------- | -------- |
On■Time Delivery | ±10%

Budget Variance | ±5%

Stakeholder Satisfaction | >4.5/5

Defect Rate | <3%

Team Satisfaction | >4.2/5

5.0 Risk Management
5.1 Risk Assessment Matrix
Probability | Low Impact | Medium Impact | High Impact

| ------------- | ------------ | --------------- | ------------- |
High | Monitor | Mitigate | Avoid

Medium | Accept | Mitigate | Transfer

Low | Accept | Monitor | Mitigate

5.2 Required Risk Reviews
• Initial risk assessment: Project start.
• Weekly: Review top 5 risks.
• Monthly: Full risk reassessment.
• Quarterly: Portfolio risk review.

6.0 Quality Assurance
6.1 Quality Gates
• Code Review: All pull requests.
• Unit Tests: >85% coverage.
• Integration Tests: Critical paths.
• UAT: Minimum 5 users (including law enforcement end■users where applicable).
• Performance Testing: Load test at 3x capacity.

6.2 Definition of Done
• Code complete.
• Tests passing.
• Documentation updated.
• Peer reviewed.
• Product Owner approved.
• AI model bias assessment completed (for AI projects).